

Bayesian Joint Quantile Trend Filtering via Proximal MCMC with Sequential Gibbs Sampling

Felipe Bedoya*

Meng Li[†]

Eric C. Chi[‡]

July 16, 2026

Abstract

We propose a fully Bayesian framework for joint quantile trend filtering of time series data. The closest existing approach is Onizuka et al. (2024), who introduced the first Bayesian quantile trend filtering method via shrinkage priors on the trend differences and a data-augmented Gibbs sampler under the asymmetric Laplace likelihood. While their method provides genuine posterior inference for individual quantile levels, it estimates each quantile independently—offering no non-crossing guarantee—and exhibits substantially degraded coverage at extreme probability levels, a limitation the authors document but do not resolve. Our framework addresses both gaps. We place independent Gamma priors on the per-quantile regularization parameters, yielding closed-form conjugate updates that allow the degree of smoothness for each quantile to be learned adaptively from the data rather than selected via an external criterion.

The core algorithmic contribution is a sequential Gibbs sampler in which each quantile trend is updated in turn via a single Proximal Metropolis-Adjusted Langevin Algorithm (P-MALA) step (Pereyra, 2016), with the non-crossing constraint enforced through exact box projections embedded in a custom ADMM proximal operator. The sequential structure reduces the joint constrained problem to a sequence of single-quantile problems, each solved with exact closed-form subproblem updates, and allows per-quantile step sizes that adapt to the geometry of each conditional posterior.

A key methodological contribution is a sentinel quantile strategy for stable estimation of extreme quantiles. We demonstrate that gradient-based samplers degenerate at extreme probability levels due to the near-constant gradient of the asymmetric Laplace likelihood, causing the Markov chain to drift. By including auxiliary sentinel quantiles beyond the target range, the sequential box constraint structure naturally confines the target quantile within a dynamically updating interval, preventing drift while preserving exploration.

Simulation studies following the design of Onizuka et al. (2024)—spanning piecewise constant and varying smoothness signals under Gaussian, Beta, and mixed normal

*Department of Statistics, Rice University. fb31@rice.edu

[†]Department of Statistics, Rice University. meng@rice.edu

[‡]School of Statistics, University of Minnesota. echi@umn.edu

noise— demonstrate that the proposed method achieves accurate, non-crossing quantile trend estimates across a wide range of probability levels, with calibrated posterior uncertainty and healthy sampler acceptance rates throughout. The method provides the uncertainty quantification absent from frequentist alternatives (Brantley et al., 2020) while matching or improving upon their point estimation accuracy, and directly resolves the extreme-quantile instability identified as an open problem in Onizuka et al. (2024).

Keywords: quantile trend filtering, Bayesian inference, proximal MCMC, ADMM, non-crossing quantiles, asymmetric Laplace distribution, Gibbs sampling, extreme quantiles.

Contents

1	Introduction	3
2	Methodology	6
2.1	Data Model and Likelihood	6
2.2	Individual Quantile Sampler	6
2.3	Joint Quantile Sampler	10
2.4	ADMM Proximal Operator for the Constrained Subproblem	11
2.5	Sentinel Quantile Strategy for Extreme Quantile Estimation	12
2.6	Summary of the Algorithm	13
3	Simulation	13
3.1	Individual Sampler	15
3.1.1	Piecewise Constant	15
3.1.2	Varying Smoothness	15
3.1.3	Smooth Function	19
3.2	Joint Sampler for Extreme Quantiles	21
3.2.1	Piecewise Constant	21
3.2.2	Varying Smoothness	22
3.2.3	Smooth Function	24
4	Discussion	28

1 Introduction

The estimation of a time-varying conditional quantile from a single observed time series is a fundamental problem in nonparametric statistics, arising in environmental monitoring, biomedical signal processing, finance, and astrophysics, among other fields. Unlike mean regression, quantile estimation provides a direct view of the upper or lower tails of the response distribution, which are often of greater scientific interest than the central tendency. When the underlying trend is believed to be piecewise polynomial—constant level shifts, piecewise linear behavior, or piecewise quadratic curvature—trend filtering provides a flexible and computationally tractable framework, producing estimates that adapt automatically to abrupt changes while remaining smooth in regions of slow variation.

The frequentist approach to quantile trend filtering was developed by Brantley et al. (2020), who combined the asymmetric Laplace check loss of Koenker and Bassett Jr (1978) with the discrete difference penalty of Tibshirani (2014) and solved the resulting nonsmooth convex program via the alternating direction method of multipliers (ADMM). Their estimator is fast, scales well to large signals, and inherits the locally adaptive properties of ℓ_1 trend filtering. Theoretical properties of the resulting estimator, including minimax risk bounds over piecewise polynomial signal classes, have been established by Madrid Padilla and Chatterjee (2022). The procedure is, however, an optimization solver: it returns a single point estimate of the trend at a prespecified penalty level, with no associated measure of uncertainty. Credible or confidence intervals must be obtained externally—typically through the bootstrap, which does not address the regularization parameter selection problem and inflates the computational burden by a factor equal to the number of resamples. The scale of the asymmetric Laplace likelihood is, without loss of generality, fixed at unity in the ADMM formulation, since it does not affect the location of the optimum; this convention disappears as soon as one wishes to treat the model as a probabilistic object.

These limitations motivate a fully Bayesian treatment of the same problem. A Bayesian formulation of quantile trend filtering yields a posterior distribution over the trend, providing credible intervals and other coherent uncertainty summaries as a direct byproduct of inference, while allowing the regularization parameter to be inferred from the data rather than tuned externally. Related progress on Bayesian quantile regression in the additive and spline settings has been made by Fasiolo et al. (2021), though their approach targets smooth additive models rather than piecewise polynomial trends of arbitrary order. The analogous trajectory for mean trend filtering has been traced by Heng, Zhou, and Chi, who placed the ℓ_1 trend filtering problem of Tibshirani (2014) in a Bayesian framework using proximal Markov chain Monte Carlo techniques (Pereyra, 2016) to handle the nondifferentiable prior and obtain posterior samples along with automatic, data-driven calibration of the smoothing parameter. The quantile setting inherits the same motivation but introduces a substantive additional difficulty: the asymmetric Laplace likelihood is itself nonsmooth, so a Bayesian quantile trend filtering sampler must contend with two sources of nondifferentiability rather than one.

A closely related Bayesian quantile trend filtering procedure is the Gibbs sampler of Onizuka et al. (2024), which exploits the scale-mixture representation of the asymmetric Laplace distribution to construct conjugate full conditionals under either a Bayesian Lasso or a horseshoe prior placed locally on each coordinate of the difference vector $D\theta$. The

resulting sampler is principled, provides genuine posterior inference, and achieves strong empirical performance, particularly on piecewise constant signals where the per-coefficient horseshoe prior delivers high local adaptivity. Two structural limitations, however, restrict its practical reach. First, the trend update requires sampling a multivariate Gaussian whose precision matrix is $D^\top \text{diag}(w)^{-1} D + \text{diag}(c)$, where the diagonal weights w are themselves random variables resampled at every Gibbs iteration. The matrix therefore changes from one iteration to the next, foreclosing any possibility of caching its factorization, and its inversion incurs the full $O(n^3)$ dense linear algebra cost at every step, making the exact MCMC sampler slow even at moderate sample sizes. Second, the variational Bayes (VB) approximation proposed as a faster alternative relies on ratios of modified Bessel functions of high order, which become numerically unstable at moderate sample sizes; and even were these numerical issues resolved, mean-field variational approximations are well known to systematically underestimate posterior variance, undermining the very inferential guarantees that motivate the Bayesian framework.

The present paper develops a Bayesian quantile trend filtering procedure that retains the scalability and structural advantages of the frequentist ADMM solver while delivering full posterior inference. The model is the standard one: an asymmetric Laplace likelihood at the target quantile level τ and a global Laplace prior on the $(k + 1)$ -th order differences of the trend, equivalent in posterior mode to the Brantley et al. (2020) estimator. The contribution lies in the sampler. We construct a three-block Gibbs scheme in which the regularization parameter λ is updated from a conjugate Gamma full conditional, the asymmetric Laplace scale σ —treated as an unknown parameter rather than fixed at unity, in contrast to the optimization formulation—is updated from a conjugate inverse-Gamma full conditional, and the trend θ is updated via a single step of proximal Metropolis-adjusted Langevin algorithm (P-MALA) (Pereyra, 2016). The proximal map required by P-MALA is itself a constrained ℓ_1 trend filtering problem, which we solve by a small number of ADMM iterations whose linear system has a fixed coefficient matrix that depends only on the deterministic step sizes and the difference operator. Because this matrix does not change between Gibbs iterations and is sparse and banded, it admits a Cholesky factorization that is computed once and reused thereafter, with backsolve cost linear in n . The sampler is therefore fully Bayesian, provides automatic and self-calibrating updates for both the regularization strength and the noise scale, and exhibits per-iteration complexity that is substantially lower than the dense Gibbs sampler of Onizuka et al. (2024).

The framework extends naturally to joint inference across multiple quantile levels under the standard non-crossing constraint. The single-quantile P-MALA step is replaced by a sequential update in which the proximal map is solved under box constraints determined by the most recent values of the neighboring quantile trends. The required modification to the ADMM solver introduces a single auxiliary variable and an elementwise projection step, leaving its overall structure and computational profile intact. This joint construction, however, also exposes a difficulty intrinsic to gradient-based samplers under the asymmetric Laplace likelihood at extreme quantile levels. For τ close to zero or one, nearly all residuals share the same sign and the gradient of the smoothed check loss approaches a constant vector, so the P-MALA proposal carries negligible directional information and the chain develops a persistent drift away from the data. The locally adaptive prior of Onizuka et al. (2024) partially mitigates this pathology, and their sampler produces degraded but not catastrophic

estimates at extreme quantile levels. We instead resolve it through a strategy that exploits the box-constraint structure of the joint sampler: we augment the set of inferential quantile levels with auxiliary *sentinel* quantiles placed beyond the target range, whose role is purely structural. The sentinels are sampled as ordinary members of the joint posterior and provide a dynamically updating upper or lower bound on the extreme target quantile through the non-crossing constraint, confining its exploration to a statistically sensible region at every iteration. The mechanism is computationally cheap and, in our simulation study, allows the proposed sampler to match or slightly improve upon the extreme-quantile accuracy of Onizuka et al. (2024) while using a globally penalized prior that is structurally simpler than the per-coefficient horseshoe.

The main contributions of this work are as follows.

1. We introduce a Bayesian quantile trend filtering sampler in which the trend, the regularization parameter, and the asymmetric Laplace scale are all inferred jointly from conjugate or nearly conjugate full conditionals, embedding a proximal MALA step for the nondifferentiable component of the posterior. The scale parameter is treated as an unknown, breaking with the optimization convention of fixing it at unity and allowing the effective likelihood temperature to adapt to the data.
2. We exploit the deterministic structure of the proximal subproblem to obtain a sampler whose per-iteration cost is linear in the sample size, in contrast to the cubic per-iteration cost of the existing Bayesian competitor. The coefficient matrix of the inner ADMM linear system is sparse, banded, and constant within and across Gibbs iterations, so its factorization is computed once and reused, eliminating the dominant computational bottleneck of the locally adaptive Gibbs approach of Onizuka et al. (2024).
3. We extend the sampler to the joint estimation of multiple quantile levels under non-crossing constraints, incorporating the constraints exactly through a box-projection step inside the inner ADMM solver rather than through approximate or penalty-based enforcement.
4. We introduce the sentinel quantile strategy, a structurally minimal device that resolves the gradient degeneracy of proximal MALA at extreme quantile levels and recovers competitive accuracy at the tails of the response distribution. The mechanism is generic and applies to any gradient-based sampler for quantile trend filtering, not only the one developed here.
5. Through a comprehensive simulation study spanning four signal structures, three noise distributions, three difference orders, and both interior and extreme quantile levels, we demonstrate that the proposed sampler delivers accuracy comparable to or better than the competitive Bayesian approach of Onizuka et al. (2024), with substantially reduced computational cost and proper posterior uncertainty quantification, and that it does so using a single global penalty parameter rather than the per-coordinate local shrinkage employed by competing priors, which simplifies the prior specification and the resulting sampler.

The remainder of the paper is organized as follows. Section 2 develops the methodology, presenting the data model and likelihood, the individual-quantile sampler with its three conjugate and proximal updates, the joint sampler with non-crossing constraints and the associated ADMM proximal operator, and the sentinel quantile strategy. Section 3 reports the simulation study comparing the proposed sampler with the relevant frequentist and Bayesian competitors across interior and extreme quantile levels. Section 4 discusses the results, limitations, and avenues for extension.

2 Methodology

2.1 Data Model and Likelihood

Let $\mathbf{y} = (y_1, \dots, y_n)^\top \in \mathbb{R}^n$ denote an observed time series. We model the data as

$$y_i = \theta_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (1)$$

where $\theta = (\theta_1, \dots, \theta_n)^\top$ is an unknown smooth trend and ε_i are independent errors. For a given quantile level $\tau \in (0, 1)$, we model the errors as following an asymmetric Laplace distribution (ALD). The ALD with location zero, scale $\sigma > 0$, and skewness parameter τ has density

$$f(u \mid \tau, \sigma) = \frac{\tau(1 - \tau)}{\sigma} \exp\left\{-\frac{\rho_\tau(u)}{\sigma}\right\}, \quad (2)$$

where $\rho_\tau(u) = u(\tau - \mathbb{1}(u < 0))$ is the asymmetric check function. This choice of likelihood connects the Bayesian framework directly to frequentist quantile regression: the maximum a posteriori (MAP) estimate of θ under a flat prior coincides with the solution to the standard quantile regression problem. Rather than fixing σ , we treat it as an unknown parameter and assign it a prior, so that the scale of the likelihood adapts to the data within the Gibbs scheme.

The log-likelihood for θ given $\mathbf{y}$ at quantile level τ and scale σ is

$$\ell(\theta \mid \mathbf{y}, \tau, \sigma) = n \log \left[\frac{\tau(1 - \tau)}{\sigma} \right] - \frac{1}{\sigma} \sum_{i=1}^n \rho_\tau(y_i - \theta_i).$$

2.2 Individual Quantile Sampler

We first describe the sampler that targets a single quantile level τ . The joint sampler of Section 2.3 builds directly on this construction.

Prior specification. To encourage smoothness of the estimated trend, we place a penalty on the $(k + 1)$ -th order discrete differences of θ . Let $D = D^{(k+1)} \in \mathbb{R}^{m \times n}$ denote the discrete difference operator of order $k + 1$, where $m = n - k - 1$ is its row rank. We use the hierarchical prior

$$p(\theta \mid \lambda) \propto \exp\{-\lambda \|D\theta\|_1\},$$

which corresponds to placing independent Laplace distributions on each of the m entries $(D\theta)_j$, all sharing the common scale parameter λ . This prior, combined with the ALD likelihood, yields a posterior whose MAP is the frequentist quantile trend filtering estimator of Brantley et al. (2020).

Rather than treating λ as a fixed tuning parameter selected by an external criterion, we place a Gamma hyperprior on λ ,

$$\lambda \sim \text{Gamma}(a_0, b_0),$$

with shape $a_0 > 0$ and rate $b_0 > 0$. For the ALD scale, we place an inverse-Gamma prior

$$\sigma \sim \text{IG}(a_\sigma, b_\sigma),$$

with shape $a_\sigma > 0$ and rate $b_\sigma > 0$. In our implementation we set $a_\sigma = b_\sigma = 0.1$, a weakly informative choice that lets the data dominate the posterior for σ . Both hyperpriors are conjugate given θ , yielding closed-form full conditionals that are sampled as part of the inference.

Hyperparameter scaling by difference order. The conjugate full conditional for λ is $\text{Gamma}(a_0 + m, b_0 + S_D)$ with posterior mean $(a_0 + m)/(b_0 + S_D)$, where $S_D = \|D\theta\|_1$. Because $m = n - k - 1$ grows linearly with n , the appropriate scaling of (a_0, b_0) depends on the difference order k , and the qualitative consequences of misspecification differ markedly across orders. The guiding principle is that the posterior mean of λ should scale appropriately with the complexity of the signal class being fit.

For $k = 0$, the estimator is piecewise constant and an unbounded posterior mean of λ obliterates the jump structure of the signal by shrinking the first differences of θ toward zero. The required scaling is $a_0 = c_1 n$ and $b_0 = n$, yielding posterior mean approximately $c_1 + 1$ independent of n , with prior variance c_1/n that shrinks with sample size. We use $c_1 = 1$ as the default and recommend the posterior mode as the point estimate. Alternative scalings of the form $a_0 = 1$, $b_0 = c_2 n^\alpha$ with $\alpha < 1$ produce a posterior mean of order $n^{1-\alpha}$, which diverges and is not viable for piecewise constant signals.

For $k = 1$, the estimator is piecewise linear, and slow growth of λ with n is appropriate: more data warrant stronger regularization, but excessive penalization suppresses local features such as spikes or sharp curvature transitions. We set $a_0 = 1$ and $b_0 = c_2 \sqrt{n}$, giving posterior mean approximately $\sqrt{n}/c_2$. The shape parameter $a_0 = 1$ is nearly noninformative, so the posterior mean of λ is driven by m and the sufficient statistic S_D . We use $c_2 = 1$ as the default, with the range $[0.5, 2]$ providing a reasonable operating window, and recommend the posterior mean as the point estimate.

For $k = 2$, the estimator is piecewise quadratic and the sufficient statistic $S_D = \|D^{(3)}\theta\|_1$ remains small regardless of n for signals well approximated by piecewise quadratics. Linear growth of λ in n is then appropriate: we set $a_0 = cn$ and $b_0 = 1$, yielding posterior mean approximately $(c + 1)n$. We use $c = 1$ as the default and the posterior mean as the point estimate. This order exhibits a degree of robustness that lower orders do not share: misspecification of c produces gradual degradation in mean squared error rather than the catastrophic structural collapse observed for $k = 0$.

We adopt $k = 2$ as the default difference order in the absence of prior knowledge about signal structure. It does not require commitment to a specific signal class, exhibits the lowest sensitivity to hyperparameter misspecification among the three orders, and degrades gracefully under it. Lower orders may outperform $k = 2$ when the signal class matches the estimator structure exactly and the hyperparameters are well-tuned, but the gain is modest relative to the additional tuning effort required.

Posterior. The joint posterior over $(\theta, \lambda, \sigma)$ is

$$p(\theta, \lambda, \sigma \mid \mathbf{y}) \propto \sigma^{-n} \exp \left\{ -\frac{1}{\sigma} \sum_{i=1}^n \rho_{\tau}(y_i - \theta_i) - \lambda \|D\theta\|_1 \right\} \lambda^{a_0-1} e^{-b_0\lambda} \sigma^{-a_{\sigma}-1} e^{-b_{\sigma}/\sigma}. \quad (3)$$

Direct sampling from (3) is intractable due to the non-smooth ALD likelihood and the ℓ_1 penalty. We construct a three-block Gibbs sampler that alternates between sampling $\theta \mid \lambda, \sigma, \mathbf{y}$ via proximal MALA, $\lambda \mid \theta, \mathbf{y}$ from its Gamma conjugate update, and $\sigma \mid \theta, \mathbf{y}$ from its inverse-Gamma conjugate update.

Sampling the trend via P-MALA. Sampling $\theta \mid \lambda, \sigma, \mathbf{y}$ requires drawing from a distribution with a non-smooth log-density. The P-MALA algorithm of Pereyra (2016) handles this by incorporating the proximal operator of the non-smooth component into the Langevin proposal, yielding proposals that respect the geometry of the posterior. The negative log-likelihood contribution from the ALD is $f(\theta) = \sigma^{-1} \sum_{i=1}^n \rho_{\tau}(y_i - \theta_i)$, which depends on σ both as a multiplicative scale and through the residual magnitude. To compute the gradient required by P-MALA, we use a smooth approximation of the check function evaluated on the standardized residual $u_i = (y_i - \theta_i)/\sigma$,

$$\tilde{\rho}_{\tau}(u, \varepsilon) = \tau u + \varepsilon \log(1 + e^{-u/\varepsilon}),$$

for a small smoothing parameter $\varepsilon > 0$. The gradient of the smoothed negative log-likelihood $\tilde{f}(\theta) = \sigma^{-1} \sum_{i=1}^n \tilde{\rho}_{\tau}((y_i - \theta_i)/\sigma, \varepsilon)$ with respect to θ_i is

$$\nabla_{\theta_i} \tilde{f}(\theta) = -\frac{1}{\sigma} \left(\tau - \frac{1}{1 + e^{(y_i - \theta_i)/(\sigma\varepsilon)}} \right).$$

The factor σ^{-1} means that smaller values of σ produce sharper gradients and stronger drift toward the quantile, while larger values flatten the gradient; this is the mechanism by which the scale parameter actively shapes the P-MALA proposal rather than acting as an inert nuisance.

The non-smooth component of the log-posterior is $g(\theta) = \lambda \|D\theta\|_1$. Its proximal operator with step size $\gamma > 0$ is

$$\text{prox}_{\gamma g}(z) = \arg \min_{\theta \in \mathbb{R}^n} \left\{ \lambda \|D\theta\|_1 + \frac{1}{2\gamma} \|\theta - z\|_2^2 \right\},$$

which is the ℓ_1 trend filtering problem with response z and penalty $\gamma\lambda$. In the individual sampler this proximal map is unconstrained and we solve it directly using the `trendfilter`

function of the `glmgen` package (Tibshirani, 2014), which provides an efficient ADMM-based solver for the standard trend filtering problem.

Given the current state $\theta^{(t)}$, the P-MALA proposal is

$$\theta^* = \text{prox}_{\gamma g} \left(\theta^{(t)} - \delta \nabla \tilde{f}(\theta^{(t)}) + \sqrt{2\delta} \xi \right),$$

where $\xi \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$ and $\delta > 0$ is the P-MALA step size. The proposal is accepted with the Metropolis-Hastings probability $\min(1, \alpha)$, where

$$\log \alpha = \log p(\theta^* \mid \lambda, \sigma, \mathbf{y}) - \log p(\theta^{(t)} \mid \lambda, \sigma, \mathbf{y}) + \log q(\theta^{(t)} \mid \theta^*) - \log q(\theta^* \mid \theta^{(t)}),$$

and the proposal log-density is

$$\log q(\theta^* \mid \theta^{(t)}) = -\frac{1}{4\delta} \|\theta^* - \mu^{(t)}\|_2^2, \quad \mu^{(t)} = \theta^{(t)} - \delta \nabla \psi(\theta^{(t)}),$$

with $\nabla \psi(\theta) = (\theta - \text{prox}_{\gamma g}(\theta - \gamma \nabla \tilde{f}(\theta))) / \gamma$ the gradient of the Moreau envelope. The posterior log-density in the acceptance ratio uses the exact (non-smoothed) check function, evaluated with the current value of σ .

Sampling the regularization parameter. Given θ , the conditional posterior for λ follows directly from (3). Because the prior places independent Laplace distributions on each of the m entries $(D\theta)_j$ with shared scale λ , the likelihood contribution of θ to λ is

$$p(\theta \mid \lambda) \propto \lambda^m \exp\{-\lambda \|D\theta\|_1\}.$$

Combining with the Gamma hyperprior,

$$p(\lambda \mid \theta, \mathbf{y}) \propto \lambda^{a_0+m-1} \exp\{-(b_0 + \|D\theta\|_1) \lambda\},$$

which is the kernel of a Gamma distribution,

$$\lambda \mid \theta, \mathbf{y} \sim \text{Gamma}(a_0 + m, b_0 + S_D), \quad S_D = \|D\theta\|_1. \quad (4)$$

The sufficient statistic S_D measures the total variation of the current trend estimate. When the sampled θ is rough, S_D is large and the posterior for λ shifts toward larger values, increasing regularization in the next iteration; when θ is smooth, λ is drawn from a distribution with smaller mean. This self-regulating behavior is a principal advantage of the Bayesian formulation over a fixed eBIC-selected penalty.

Sampling the ALD scale. Given (θ, λ) , only the likelihood and the inverse-Gamma prior on σ contribute to its full conditional. From (3),

$$p(\sigma \mid \theta, \mathbf{y}) \propto \sigma^{-(n+a_\sigma+1)} \exp\left\{-\frac{1}{\sigma} \left(\sum_{i=1}^n \rho_\tau(y_i - \theta_i) + b_\sigma \right)\right\},$$

which is the kernel of an inverse-Gamma distribution,

$$\sigma \mid \theta, \mathbf{y} \sim \text{IG}(n + a_\sigma, S_\rho + b_\sigma), \quad S_\rho = \sum_{i=1}^n \rho_\tau(y_i - \theta_i). \quad (5)$$

The sufficient statistic S_ρ is the total check loss at the current trend estimate. When the residuals are large, S_ρ is large and the posterior for σ concentrates at larger values, which flattens the likelihood gradient at the next P-MALA step; when the trend fits the data well, σ is drawn from a distribution with smaller mean, sharpening the gradient. The three blocks of the Gibbs sampler therefore interact: σ calibrates the effective noise level seen by θ , λ calibrates the effective smoothness penalty seen by θ , and θ in turn determines the sufficient statistics for both scale parameters.

2.3 Joint Quantile Sampler

We now extend the individual sampler to target K quantile levels simultaneously. Let $\tau_1 < \tau_2 < \dots < \tau_K$ denote the quantile levels of interest. Each quantile inherits its own copy of the full parameter set from the individual construction: a trend θ_{τ_k} , a regularization parameter λ_{τ_k} , and an ALD scale σ_{τ_k} . We seek joint posterior inference on $(\theta_{\tau_1}, \dots, \theta_{\tau_K}, \lambda_{\tau_1}, \dots, \lambda_{\tau_K}, \sigma_{\tau_1}, \dots, \sigma_{\tau_K})$ subject to the non-crossing constraint

$$\theta_{\tau_1, i} \leq \theta_{\tau_2, i} \leq \dots \leq \theta_{\tau_K, i}, \quad i = 1, \dots, n.$$

The joint posterior is

$$\begin{aligned} p(\theta_{\tau_1}, \dots, \theta_{\tau_K}, \lambda_{\tau_1}, \dots, \lambda_{\tau_K}, \sigma_{\tau_1}, \dots, \sigma_{\tau_K} \mid \mathbf{y}) &\propto \prod_{k=1}^K \left[\sigma_{\tau_k}^{-(n+a_\sigma+1)} \lambda_{\tau_k}^{a_0-1} e^{-b_0 \lambda_{\tau_k}} \right. \\ &\quad \times \exp \left\{ -\frac{1}{\sigma_{\tau_k}} \sum_{i=1}^n \rho_{\tau_k}(y_i - \theta_{\tau_k, i}) - \lambda_{\tau_k} \|D\theta_{\tau_k}\|_1 - \frac{b_\sigma}{\sigma_{\tau_k}} \right\} \Big] \cdot \mathbb{1}(\mathcal{C}), \end{aligned}$$

where $\mathcal{C}$ denotes the non-crossing feasible set.

We sample from this posterior using a fixed-scan Gibbs sampler. Each full Gibbs iteration proceeds as follows.

Sequential trend updates. For $k = 1, \dots, K$, sample θ_{τ_k} via a single P-MALA step of the form described in Section 2.2, using the current values of λ_{τ_k} and σ_{τ_k} in the gradient and acceptance ratio, with the non-crossing constraint enforced as a box constraint. At step k , the constraint reduces to

$$\ell_{\tau_k, i} \leq \theta_{\tau_k, i} \leq u_{\tau_k, i}, \quad i = 1, \dots, n,$$

where $\ell_{\tau_k, i} = \theta_{\tau_{k-1}, i}$ (or $-\infty$ if $k = 1$) and $u_{\tau_k, i} = \theta_{\tau_{k+1}, i}$ (or $+\infty$ if $k = K$), using the most recently updated values of adjacent trends. The box constraint at each step is exact: it is fully determined by the current values of the neighboring quantile trends, which are updated sequentially so that the lower bound always reflects the most recent sample of $\theta_{\tau_{k-1}}$.

The presence of the box constraint changes the proximal subproblem. The unconstrained `glmgen::trendfilter` solver used in the individual sampler no longer applies, and we instead solve the constrained proximal map via ADMM; see Section 2.4.

Sequential scale and regularization updates. For $k = 1, \dots, K$, sample λ_{τ_k} and σ_{τ_k} from their conditional posteriors given θ_{τ_k} . The derivations are identical to the individual case in (4) and (5) applied quantile-by-quantile, so

$$\begin{aligned}\lambda_{\tau_k} \mid \theta_{\tau_k}, \mathbf{y} &\sim \text{Gamma}(a_0 + m, b_0 + \|D\theta_{\tau_k}\|_1), \\ \sigma_{\tau_k} \mid \theta_{\tau_k}, \mathbf{y} &\sim \text{IG}(n + a_\sigma, \sum_{i=1}^n \rho_{\tau_k}(y_i - \theta_{\tau_k,i}) + b_\sigma),\end{aligned}$$

for $k = 1, \dots, K$. Each λ_{τ_k} and σ_{τ_k} is updated independently, so that each quantile's degree of smoothing and its effective noise scale adapt to the fit of its own estimated trend.

Parameter structure. Each quantile τ_k has its own P-MALA step size δ_k , allowing the proposal to adapt to the differing curvatures of the conditional posteriors across quantile levels. The smoothing parameter ε for the check loss approximation, the proximal step size γ , and the ADMM penalty parameter ρ are shared across all quantiles. The Gamma prior hyperparameters (a_0, b_0) and the inverse-Gamma hyperparameters (a_σ, b_σ) are also shared, encoding a common prior belief about the scale of regularization and the noise level.

2.4 ADMM Proximal Operator for the Constrained Subproblem

At each Gibbs step of the joint sampler, the P-MALA proposal for θ_{τ_k} requires solving the constrained proximal problem

$$\min_{\theta \in \mathbb{R}^n} \left\{ \lambda_{\tau_k} \|D\theta\|_1 + \frac{1}{2\gamma} \|\theta - z\|_2^2 \right\} \quad \text{subject to} \quad \boldsymbol{\ell} \leq \theta \leq \mathbf{u}, \quad (6)$$

where z is the P-MALA input and $[\boldsymbol{\ell}, \mathbf{u}]$ is the box defined by the neighboring quantile trends. We solve (6) via ADMM by introducing auxiliary variables $v = D\theta$ and $w = \theta$, yielding the equivalent problem

$$\min_{\theta, v, w} \left\{ \lambda_{\tau_k} \|v\|_1 + \frac{1}{2\gamma} \|\theta - z\|_2^2 \right\} \quad \text{subject to} \quad v = D\theta, \quad w = \theta, \quad \boldsymbol{\ell} \leq w \leq \mathbf{u}.$$

The augmented Lagrangian is

$$\begin{aligned}\mathcal{L}_\rho(\theta, v, w, \alpha, \beta) &= \lambda_{\tau_k} \|v\|_1 + \frac{1}{2\gamma} \|\theta - z\|_2^2 + \alpha^\top (D\theta - v) + \frac{\rho}{2} \|D\theta - v\|_2^2 \\ &\quad + \beta^\top (\theta - w) + \frac{\rho}{2} \|\theta - w\|_2^2,\end{aligned}$$

where $\rho > 0$ is the ADMM penalty parameter and α, β are dual variables. The ADMM updates at iteration t are:

θ -update (linear system). Minimizing over θ gives

$$\left(\frac{1}{\gamma} \mathbf{I} + \rho D^\top D + \rho \mathbf{I} \right) \theta^{t+1} = \frac{1}{\gamma} z + D^\top (\rho v^t - \alpha^t) + (\rho w^t - \beta^t).$$

The system matrix $\mathbf{A} = \frac{1}{\gamma} \mathbf{I} + \rho D^\top D + \rho \mathbf{I}$ depends only on γ , ρ , and D , all of which are fixed within a Gibbs iteration. Exploiting this, we factorize $\mathbf{A}$ once at the beginning of each Gibbs step and reuse the factorization across all ADMM iterations. Combined with sparse storage of D and $D^\top D$, this constitutes the primary computational optimization of the algorithm.

v -update (soft-thresholding). Minimizing over v decouples elementwise and yields the soft-threshold operator,

$$v_j^{t+1} = \mathcal{S}_{\lambda_{\tau_k}/\rho}((D\theta^{t+1})_j + \alpha_j^t/\rho),$$

where $\mathcal{S}_\kappa(x) = \text{sign}(x) \max(|x| - \kappa, 0)$ is the soft-thresholding operator.

w -update (box projection). Minimizing over w subject to $\ell \leq w \leq \mathbf{u}$ yields the elementwise projection onto the box,

$$w_i^{t+1} = \text{clip}(\theta_i^{t+1} + \beta_i^t/\rho, \ell_i, u_i),$$

where $\text{clip}(x, \ell, u) = \max(\ell, \min(u, x))$. This is the step that enforces the non-crossing constraint: the box $[\ell, \mathbf{u}]$ is defined by the current values of the neighboring quantile trends, and the projection is exact and elementwise.

Dual updates.

$$\begin{aligned}\alpha^{t+1} &= \alpha^t + \rho(D\theta^{t+1} - v^{t+1}), \\ \beta^{t+1} &= \beta^t + \rho(\theta^{t+1} - w^{t+1}).\end{aligned}$$

All three subproblem updates have exact closed-form solutions, and the algorithm converges to the global optimum of (6) as the number of ADMM iterations grows. In practice, we run a fixed number of ADMM iterations per Gibbs step, with convergence monitored via primal and dual residuals.

2.5 Sentinel Quantile Strategy for Extreme Quantile Estimation

A fundamental difficulty in Bayesian sampling for extreme quantile trend filtering arises from the behavior of the ALD likelihood gradient. For a quantile level τ close to zero or one, nearly all residuals $y_i - \theta_i$ have the same sign, and the gradient of the smoothed check loss $\nabla \tilde{f}(\theta)$ approaches a constant vector. The P-MALA proposal then carries negligible gradient information and degenerates toward a random walk. Because the posterior at extreme quantile levels has a heavy tail in the direction away from the bulk of the data, the Markov chain occasionally accepts large moves in that direction, causing persistent upward or downward drift. For the 0.99 quantile with $n = 400$, this pathology produces large RMSE values under individual estimation, rendering the estimate essentially useless. Onizuka et al. (2024) document degraded coverage at $\tau \in \{0.05, 0.95\}$ across their full simulation suite, which spans two signal structures—piecewise constant (PC) and varying smoothness (VS)—under three noise distributions: Gaussian, Beta, and mixed normal.

The sentinel quantile strategy resolves this by exploiting the sequential box constraint structure of the joint sampler. We augment the set of quantiles of inferential interest with auxiliary sentinel quantiles placed beyond the target range. The sentinels are treated as full members of the joint posterior: they are sampled by the same Gibbs scheme, receive their own regularization and scale parameters, and their estimates are computable. However they are not primary targets of inference; their role is structural.

Concretely, to estimate the 0.99 quantile stably, we include sentinel quantiles at 0.995 and 0.999. The 0.995 sentinel provides a dynamic upper wall for the 0.99 quantile through the box constraint $\theta_{0.99,i} \leq \theta_{0.995,i}$, and is itself controlled from above by the 0.999 sentinel through $\theta_{0.995,i} \leq \theta_{0.999,i}$. A stable scaffold of lower quantiles (e.g., 0.3, 0.5, 0.7, 0.9, 0.95, 0.97, 0.98) anchors the structure from below, preventing downward drift of the 0.99 quantile. The result is that the 0.99 quantile is sandwiched within a dynamically updating interval at every Gibbs iteration, which confines its exploration to a statistically sensible region without imposing any fixed constraint on its value. The strategy is computationally free in the sense that adding sentinel quantiles increases the number of sequential Gibbs steps by the number of sentinels, which is small, without changing the per-step cost.

2.6 Summary of the Algorithm

Algorithm 1 summarizes the full sequential Gibbs sampler for joint quantile trend filtering.

Algorithm 1 Sequential Gibbs Sampler for Joint Bayesian Quantile Trend Filtering

Require: Data $\mathbf{y}$, quantile levels $\tau_1 < \dots < \tau_K$ (including sentinels), difference order k , prior hyperparameters $a_0, b_0, a_\sigma, b_\sigma$, ADMM iterations T_{admm} , Gibbs iterations T_{gibbs} , burn-in T_{burn} .

- 1: Initialize $\theta_{\tau_k}^{(0)}, \lambda_{\tau_k}^{(0)}, \sigma_{\tau_k}^{(0)}$ for $k = 1, \dots, K$
- 2: **for** $t = 1, \dots, T_{\text{gibbs}}$ **do**
- 3: **for** $k = 1, \dots, K$ **do** ▷ Sequential trend updates
- 4: Set box $[\ell_k, \mathbf{u}_k]$ using current $\theta_{\tau_{k-1}}^{(\cdot)}$ and $\theta_{\tau_{k+1}}^{(t-1)}$
- 5: Compute gradient $\nabla \tilde{f}_k(\theta_{\tau_k}^{(t-1)})$ using $\sigma_{\tau_k}^{(t-1)}$
- 6: Form proposal input $z = \theta_{\tau_k}^{(t-1)} - \delta_k \nabla \tilde{f}_k + \sqrt{2\delta_k} \xi, \xi \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 7: Solve (6) via ADMM (Algorithm 2) to get θ^*
- 8: Accept/reject θ^* via Metropolis-Hastings using $(\lambda_{\tau_k}^{(t-1)}, \sigma_{\tau_k}^{(t-1)})$
- 9: Set $\theta_{\tau_k}^{(t)} \leftarrow$ accepted state
- 10: **end for**
- 11: **for** $k = 1, \dots, K$ **do** ▷ Sequential regularization updates
- 12: Sample $\lambda_{\tau_k}^{(t)} \sim \text{Gamma}(a_0 + m, b_0 + \left\| D\theta_{\tau_k}^{(t)} \right\|_1)$
- 13: **end for**
- 14: **for** $k = 1, \dots, K$ **do** ▷ Sequential scale updates
- 15: Sample $\sigma_{\tau_k}^{(t)} \sim \text{IG}(n + a_\sigma, \sum_{i=1}^n \rho_{\tau_k}(y_i - \theta_{\tau_k,i}^{(t)}) + b_\sigma)$
- 16: **end for**
- 17: **end for**
- 18: **return** posterior samples $\{\theta_{\tau_k}^{(t)}, \lambda_{\tau_k}^{(t)}, \sigma_{\tau_k}^{(t)}\}_{t > T_{\text{burn}}}$ for $k = 1, \dots, K$

3 Simulation

We evaluate the proposed samplers on three test signals of increasing smoothness, each observed under three error distributions, across a common grid of quantile levels. Throughout

Algorithm 2 ADMM for Constrained Proximal Subproblem

Require: Input z , box $[\ell, \mathbf{u}]$, λ , γ , ρ , D , iterations T_{admm}

- 1: Factorize $\mathbf{A} = \frac{1}{\gamma}\mathbf{I} + \rho D^T D + \rho \mathbf{I}$ ▷ Once per Gibbs step
 - 2: Initialize $v^0, w^0, \alpha^0, \beta^0$
 - 3: **for** $t = 0, \dots, T_{\text{admm}} - 1$ **do**
 - 4: $\theta^{t+1} \leftarrow \mathbf{A}^{-1} \left(\frac{1}{\gamma}z + D^\top(\rho v^t - \alpha^t) + \rho w^t - \beta^t \right)$
 - 5: $v^{t+1} \leftarrow \mathcal{S}_{\lambda/\rho}(D\theta^{t+1} + \alpha^t/\rho)$
 - 6: $w^{t+1} \leftarrow \text{clip}(\theta^{t+1} + \beta^t/\rho, \ell, \mathbf{u})$
 - 7: $\alpha^{t+1} \leftarrow \alpha^t + \rho(D\theta^{t+1} - v^{t+1})$
 - 8: $\beta^{t+1} \leftarrow \beta^t + \rho(\theta^{t+1} - w^{t+1})$
 - 9: **end for**
 - 10: **return** $\theta^{T_{\text{admm}}}$
-

we take $n = 400$ equally spaced design points $x_i = i$, $i = 1, \dots, n$, and write $\tilde{x}_i = i/n \in (0, 1]$ for the rescaled index. Every configuration is replicated over 100 independent data sets, and all reported figures are averages over these replications. The individual sampler is run for 6000 Gibbs iterations with a burn-in of 1000, and its point estimate is the posterior mean of θ unless stated otherwise. Performance is measured by mean squared error (MSE) and mean absolute deviation (MAD) of the point estimate against the true conditional quantile Q_τ , and by the mean credible interval width (MCIW) of the 95% pointwise posterior intervals.

Baseline signals. The mean function f is one of the following, indexed by the difference order k at which it is fit:

$$\begin{aligned} \text{(a) Piecewise constant } (k = 0) : \quad f(\tilde{x}) &= 2.5\mathbb{I}\{\tilde{x} \in (0, 0.2]\} + 1.0\mathbb{I}\{\tilde{x} \in (0.2, 0.4]\} \\ &\quad + 3.5\mathbb{I}\{\tilde{x} \in (0.4, 0.6]\} + 1.5\mathbb{I}\{\tilde{x} \in (0.6, 1]\}, \end{aligned} \quad (7)$$

$$\text{(b) Varying smoothness } (k = 1) : \quad f(\tilde{x}) = 2 + \sin(4\tilde{x} - 2) + 2 \exp(-30(4\tilde{x} - 2)^2), \quad (8)$$

$$\text{(c) Smooth } (k = 2) : \quad f(\tilde{x}) = \sin(2\pi\tilde{x}). \quad (9)$$

Signal (a) is piecewise constant with three change points, signal (b) combines a slowly varying sinusoid with a sharply localized bump, and signal (c) is globally smooth.

Noise distributions. Observations are generated as $y_i = f(\tilde{x}_i) + \varepsilon_i$, with a design-dependent error ε_i drawn from one of:

- *Gaussian (heteroscedastic):* $\varepsilon_i \sim \mathcal{N}(0, s(\tilde{x}_i)^2)$ with $s(\tilde{x}) = (1 + \tilde{x}^2)/4$, so that $Q_\tau(\tilde{x}) = f(\tilde{x}) + s(\tilde{x})\Phi^{-1}(\tau)$.
- *Beta:* $\varepsilon_i \sim \text{Beta}(1, 11 - 10\tilde{x}_i)$, so that $Q_\tau(\tilde{x}) = f(\tilde{x}) + F_{\text{Beta}(1, 11-10\tilde{x})}^{-1}(\tau)$; the error is right-skewed and its shape varies with the design.
- *Mixed normal (location mixture):* $\varepsilon_i \sim \tilde{x}_i \mathcal{N}(-0.2, 0.5) + (1 - \tilde{x}_i) \mathcal{N}(0.2, 0.5)$, a two-component mixture whose mixing weight drifts with the design. The true quantile

$Q_\tau(\tilde{x}) = f(\tilde{x}) + G_{\tilde{x}}^{-1}(\tau)$ is obtained by inverting the mixture CDF $G_{\tilde{x}}(t) = \tilde{x} \Phi((t + 0.2)/\sqrt{0.5}) + (1 - \tilde{x}) \Phi((t - 0.2)/\sqrt{0.5})$.

The Gaussian case is symmetric but heteroscedastic, the Beta case is skewed, and the mixed normal case is the most adversarial to the asymmetric Laplace working likelihood because its shape—not merely its scale—changes across the domain.

3.1 Individual Sampler

This subsection reports the single-quantile sampler at the interior levels $\tau \in \{0.05, 0.25, 0.5, 0.75, 0.95\}$, against the two Onizuka variants (Laplace and Horseshoe), both run in MCMC mode.

3.1.1 Piecewise Constant

On the piecewise constant signal the horseshoe prior of Onizuka is the strongest competitor: its local adaptivity is well matched to a mean function that is flat almost everywhere with a few abrupt jumps, and it attains the lowest MSE and MAD across the interior of the Gaussian and mixed normal designs. The individual sampler improves uniformly on the Laplace variant and is most competitive at the near-tail levels $\tau \in \{0.05, 0.95\}$, where under the mixed normal design it records the lowest error of the three methods; it also leads across the Beta design at the lower levels. Away from the tails under Gaussian and mixed normal noise, however, the individual sampler does not match the horseshoe, and these tables should not be read as a claim of dominance on piecewise constant structure. Interval widths (Table 3) are reported only as a descriptive summary of posterior dispersion; under a misspecified asymmetric Laplace likelihood the frequentist coverage of these intervals is not guaranteed, and empirical coverage remains the decisive diagnostic.

3.1.2 Varying Smoothness

On the varying smoothness signal the individual sampler is broadly competitive with the horseshoe and improves on the Laplace variant across most levels and noise distributions. Under Gaussian noise it sits between the two Onizuka variants and overtakes the horseshoe at the upper level $\tau = 0.95$; under mixed normal noise the horseshoe retains a modest edge while the individual sampler remains ahead of the Laplace variant throughout. The exception is the lower level $\tau = 0.05$ under Beta noise, where the individual sampler records an MSE of 0.0696 (Table 4) that is an order of magnitude larger than its values at the adjacent levels and out of line with its behaviour on the same signal–quantile combination under the other error distributions; this entry is an outlier rather than a representative summary and is under investigation. Setting that entry aside, at the remaining Beta levels the individual sampler attains the lowest MSE and MAD among the three methods. As in the piecewise constant case, interval widths are descriptive of posterior dispersion only, and empirical coverage is the decisive measure of calibration.

Table 1: Mean squared error (MSE) for the piecewise constant signal ($k = 0$) at interior quantile levels, across noise distributions.

	MSE				
(I) Gauss	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0248	0.0043	0.0027	0.0044	0.0246
Onizuka (Laplace)	0.0570	0.0257	0.0201	0.0254	0.0587
Individual Sampler	0.0426	0.0161	0.0127	0.0153	0.0398
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0146	0.0020	0.0018	0.0033	0.0103
Onizuka (Laplace)	0.0085	0.0048	0.0063	0.0112	0.0169
Individual Sampler	0.0024	0.0015	0.0027	0.0062	0.0093
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.1273	0.0256	0.0190	0.0228	0.1230
Onizuka (Laplace)	0.2679	0.0812	0.0655	0.0793	0.2754
Individual Sampler	0.1089	0.0813	0.0725	0.0792	0.1051

Table 2: Mean absolute deviation (MAD) for the piecewise constant signal ($k = 0$) at interior quantile levels, across noise distributions.

	MAD				
(I) Gauss	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.1149	0.0393	0.0344	0.0411	0.1151
Onizuka (Laplace)	0.1894	0.1212	0.1069	0.1213	0.1928
Individual Sampler	0.1300	0.0867	0.0779	0.0832	0.1250
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0680	0.0161	0.0240	0.0353	0.0751
Onizuka (Laplace)	0.0752	0.0466	0.0564	0.0768	0.0969
Individual Sampler	0.0287	0.0225	0.0337	0.0512	0.0669
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.2802	0.0944	0.0837	0.0945	0.2751
Onizuka (Laplace)	0.4328	0.2189	0.1937	0.2162	0.4417
Individual Sampler	0.2440	0.1981	0.1876	0.1949	0.2380

Table 3: Mean credible interval width (MCIW) for the piecewise constant signal ($k = 0$) at interior quantile levels, across noise distributions. Width is a descriptive summary of posterior dispersion and should be read alongside empirical coverage.

	MCIW				
(I) Gauss	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.4277	0.2085	0.1908	0.2055	0.4269
Onizuka (Laplace)	0.6691	0.6326	0.6139	0.6343	0.6684
Individual Sampler	0.2796	0.3596	0.3542	0.3603	0.2811
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0593	0.0758	0.1105	0.1519	0.2836
Onizuka (Laplace)	0.3029	0.2756	0.3081	0.3353	0.4031
Individual Sampler	0.0111	0.0940	0.1164	0.1407	0.0569
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.9432	0.4489	0.4180	0.4372	0.9177
Onizuka (Laplace)	1.2988	1.1193	1.0561	1.1161	1.3020
Individual Sampler	0.6274	0.5738	0.5703	0.5710	0.6341

Table 4: Mean squared error (MSE) for the varying smoothness signal ($k = 1$) at interior quantile levels, across noise distributions. The Beta entry at $\tau = 0.05$ is anomalous (see text).

	MSE				
(I) Gauss	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0165	0.0065	0.0055	0.0063	0.0155
Onizuka (Laplace)	0.0276	0.0099	0.0082	0.0095	0.0262
Individual Sampler	0.0231	0.0080	0.0066	0.0069	0.0149
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0146	0.0020	0.0018	0.0033	0.0103
Onizuka (Laplace)	0.0085	0.0048	0.0063	0.0112	0.0169
Individual Sampler	0.0696	0.0012	0.0018	0.0034	0.0067
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0575	0.0262	0.0218	0.0242	0.0576
Onizuka (Laplace)	0.1124	0.0366	0.0311	0.0346	0.1119
Individual Sampler	0.0778	0.0319	0.0280	0.0318	0.0756

Table 5: Mean absolute deviation (MAD) for the varying smoothness signal ($k = 1$) at interior quantile levels, across noise distributions.

	MAD				
	0.05	0.25	0.5	0.75	0.95
(I) Gauss					
Onizuka (Horseshoe)	0.0950	0.0583	0.0552	0.0584	0.0920
Onizuka (Laplace)	0.1273	0.0756	0.0692	0.0748	0.1242
Individual Sampler	0.1055	0.0686	0.0617	0.0644	0.0921
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0680	0.0161	0.0240	0.0353	0.0751
Onizuka (Laplace)	0.0752	0.0466	0.0564	0.0768	0.0969
Individual Sampler	0.0823	0.0216	0.0292	0.0406	0.0610
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.1804	0.1212	0.1114	0.1173	0.1822
Onizuka (Laplace)	0.2639	0.1479	0.1368	0.1445	0.2658
Individual Sampler	0.2197	0.1412	0.1304	0.1396	0.2164

Table 6: Mean credible interval width (MCIW) for the varying smoothness signal ($k = 1$) at interior quantile levels, across noise distributions. Width is a descriptive summary of posterior dispersion and should be read alongside empirical coverage.

	MCIW				
	0.05	0.25	0.5	0.75	0.95
(I) Gauss					
Onizuka (Horseshoe)	0.3007	0.2595	0.2551	0.2555	0.2928
Onizuka (Laplace)	0.3849	0.3489	0.3430	0.3460	0.3805
Individual Sampler	0.2423	0.2677	0.2692	0.2696	0.2344
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0593	0.0758	0.1105	0.1519	0.2836
Onizuka (Laplace)	0.3029	0.2756	0.3081	0.3353	0.4031
Individual Sampler	0.0974	0.0845	0.1160	0.1260	0.1051
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.5875	0.5147	0.5158	0.5177	0.5940
Onizuka (Laplace)	0.7676	0.6475	0.6369	0.6493	0.7671
Individual Sampler	0.5453	0.5757	0.5728	0.5741	0.5446

3.1.3 Smooth Function

The smooth signal is where the individual sampler is strongest. Under Gaussian and mixed normal noise it attains the lowest MSE and MAD of the three methods at every interior level (Tables 7 and 8), and under Beta noise it leads at all levels except $\tau = 0.05$, where the Laplace variant is marginally better. The advantage is intuitive: a globally smooth mean function offers no isolated features for the horseshoe’s local adaptivity to exploit, so the trend-filtering prior fit at $k = 2$ is well matched to the truth and the borrowing of strength across the whole domain pays off. The credible intervals of the individual sampler are also the narrowest of the three methods across all designs and levels (Table 9); consistent with the position taken above, this narrowing is reported descriptively and should not be interpreted as sharper uncertainty quantification until empirical coverage is established under the misspecified working likelihood.

Table 7: Mean squared error (MSE) for the smooth signal ($k = 2$) at interior quantile levels, across noise distributions.

	MSE				
(I) Gauss	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0177	0.0068	0.0056	0.0060	0.0148
Onizuka (Laplace)	0.0169	0.0070	0.0059	0.0061	0.0145
Individual Sampler	0.0099	0.0042	0.0033	0.0035	0.0083
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0005	0.0011	0.0017	0.0023	0.0052
Onizuka (Laplace)	0.0004	0.0011	0.0017	0.0023	0.0053
Individual Sampler	0.0005	0.0007	0.0010	0.0014	0.0030
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0749	0.0263	0.0225	0.0283	0.0761
Onizuka (Laplace)	0.0756	0.0262	0.0228	0.0279	0.0776
Individual Sampler	0.0527	0.0203	0.0168	0.0202	0.0487

Summary across signals. The three-block sampler of Section 2.2 regularizes θ through a single Laplace prior on $D\theta$ with one shared scale λ , and calibrates the effective noise level through a single scale σ ; both parameters admit closed-form conjugate full conditionals—Gamma for λ , inverse-Gamma for σ —driven respectively by the sufficient statistics $S_D = \|D\theta\|_1$ and $S_\rho = \sum_i \rho_\tau(y_i - \theta_i)$, so that the amount of smoothing and the assumed noise scale both self-adjust to the current fit of θ without any external tuning step. This is a substantially thinner hierarchy than the horseshoe construction of Onizuka (Horseshoe), which places a local shrinkage parameter on every one of the n coordinates of $D\theta$ in addition to a global scale, and it is this added hierarchy, not a fundamentally different fitting principle, that gives the horseshoe its local adaptivity to jumps and its edge on the piecewise constant signal. The simulation results are consistent with this trade-off. Against the Laplace variant,

Table 8: Mean absolute deviation (MAD) for the smooth signal ($k = 2$) at interior quantile levels, across noise distributions.

	MAD				
	0.05	0.25	0.5	0.75	0.95
(I) Gauss					
Onizuka (Horseshoe)	0.0993	0.0629	0.0569	0.0591	0.0928
Onizuka (Laplace)	0.0986	0.0644	0.0585	0.0604	0.0924
Individual Sampler	0.0737	0.0487	0.0436	0.0454	0.0693
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0097	0.0182	0.0266	0.0343	0.0557
Onizuka (Laplace)	0.0095	0.0189	0.0276	0.0355	0.0561
Individual Sampler	0.0116	0.0147	0.0207	0.0271	0.0433
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.2131	0.1281	0.1181	0.1319	0.2143
Onizuka (Laplace)	0.2136	0.1281	0.1194	0.1315	0.2154
Individual Sampler	0.1788	0.1114	0.1011	0.1118	0.1733

Table 9: Mean credible interval width (MCIW) for the smooth signal ($k = 2$) at interior quantile levels, across noise distributions. Width is a descriptive summary of posterior dispersion and should be read alongside empirical coverage.

	MCIW				
	0.05	0.25	0.5	0.75	0.95
(I) Gauss					
Onizuka (Horseshoe)	0.2799	0.2791	0.2783	0.2760	0.2731
Onizuka (Laplace)	0.2591	0.2649	0.2694	0.2655	0.2555
Individual Sampler	0.1886	0.1918	0.1922	0.1922	0.1866
(II) Beta	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.0592	0.0902	0.1208	0.1453	0.1551
Onizuka (Laplace)	0.0546	0.0888	0.1201	0.1420	0.1481
Individual Sampler	0.0147	0.0630	0.0840	0.0992	0.1003
(III) Mixed normal	0.05	0.25	0.5	0.75	0.95
Onizuka (Horseshoe)	0.5771	0.5873	0.5840	0.5862	0.5827
Onizuka (Laplace)	0.5357	0.5483	0.5543	0.5472	0.5523
Individual Sampler	0.4422	0.4577	0.4545	0.4543	0.4433

which shares the individual sampler’s single-scale prior but fixes λ by an external criterion rather than adapting it within the chain, the individual sampler is on par or ahead at essentially every level, signal, and noise distribution, indicating that the conjugate self-tuning of λ and σ is doing real work. Against the horseshoe, the individual sampler trails on the piecewise constant signal, where isolated jumps reward coordinate-specific shrinkage; is closely competitive on the varying smoothness signal; and surpasses the horseshoe at nearly every level on the smooth signal, where no localized structure exists for the extra hierarchy to exploit and the single global scale is the better-matched inductive bias. Because the horseshoe’s advantage is concentrated in signals with strong local heterogeneity, and because it is bought with n additional per-coordinate parameters sampled at every Gibbs iteration against the individual sampler’s two, the individual sampler offers a favorable accuracy-to-complexity trade-off: it recovers most of the horseshoe’s benefit where that benefit exists, exceeds it where it does not, and does so with a model that is easier to specify, easier to diagnose, and cheaper to update per iteration. A direct wall-clock comparison quantifying the per-iteration cost gap is left for future reporting.

Default difference order. When the smoothness of the underlying trend is not known in advance, our default recommendation is the difference order $k = 2$. A penalty on second differences fits piecewise linear trends and thereby provides a reasonable local approximation to a wide variety of quantile functions, from globally smooth curves to signals with localized features, degrading gracefully when the truth departs from the assumed order. The lower orders are more specialized: $k = 0$ is well matched only when the trend is genuinely piecewise constant, while $k = 1$ occupies an intermediate position. Accordingly, the order-specific fits reported above should be read as each order evaluated on its best-case signal, with $k = 2$ the sensible choice absent prior structural knowledge.

3.2 Joint Sampler for Extreme Quantiles

This subsection reports the joint sequential sampler at the extreme levels $\tau \in \{0.01, 0.02, 0.98, 0.99\}$, equipped with the sentinel construction described above. As before, the two Onizuka variants estimate each quantile in isolation and serve as the point of comparison.

3.2.1 Piecewise Constant

Tables 10, 11, and 12 report MSE, MAD, and MCIW for the piecewise constant signal at the extreme quantile levels, fit at difference order $k = 0$. At these levels the Joint Sequential Sampler recovers much of the ground it concedes to the horseshoe at the interior of this signal. It improves on Onizuka (Laplace) in MSE and MAD at every level and noise distribution. Against Onizuka (Horseshoe) the comparison splits by tail and by error distribution: under Beta noise the Joint Sequential Sampler attains the lowest MSE at all four levels; under mixed normal noise it leads at $\tau \in \{0.02, 0.98, 0.99\}$, with the horseshoe retaining an advantage only at the deepest lower level $\tau = 0.01$; and under Gaussian noise the horseshoe remains more accurate at the lower levels $\{0.01, 0.02\}$ while the Joint Sequential Sampler records the lowest MSE and MAD at the upper levels $\{0.98, 0.99\}$. The MAD rankings follow the same pattern, with one instructive exception at the lower Beta tail: there the horseshoe has the

smaller MAD while the Joint Sequential Sampler has the smaller MSE, indicating that the horseshoe is closer to the truth on typical replications but incurs occasional larger deviations that the squared loss penalizes more heavily. As elsewhere, the credible intervals of the Joint Sequential Sampler are far narrower than those of both competitors (Table 12); the origin and proper reading of this contraction are discussed at the end of the section.

Table 10: Mean squared error (MSE) for the piecewise constant signal at extreme quantile levels, difference order $k = 0$, across noise distributions. The Joint Sequential Sampler is the sequential joint sampler equipped with sentinel quantiles.

	MSE			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.0676	0.0544	0.0496	0.0655
Onizuka (Laplace)	0.1217	0.1024	0.1002	0.1187
Joint Sequential Sampler	0.1061	0.0826	0.0296	0.0484
(II) Beta	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.0283	0.0114	0.0164	0.0224
Onizuka (Laplace)	0.0166	0.0076	0.0325	0.0363
Joint Sequential Sampler	0.0028	0.0011	0.0110	0.0154
(III) Mixed normal	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.3788	0.2715	0.2819	0.3887
Onizuka (Laplace)	0.5896	0.4673	0.4811	0.6037
Joint Sequential Sampler	0.4092	0.2103	0.1276	0.2630

3.2.2 Varying Smoothness

Tables 13, 14, and 15 report MSE, MAD, and MCIW for the varying smoothness signal at the extreme quantile levels $\tau \in \{0.01, 0.02, 0.98, 0.99\}$, fit at difference order $k = 1$ and estimated with the joint sequential sampler under the sentinel construction described above. In point estimation the Joint Sequential Sampler improves on Onizuka (Laplace) at essentially all levels and is competitive with Onizuka (Horseshoe). The advantage is clearest under mixed normal noise, where the Joint Sequential Sampler attains the lowest MSE and MAD at every extreme level, including the upper tail $\{0.98, 0.99\}$ where the location mixture is most adversarial to the asymmetric Laplace working likelihood. Under Gaussian noise it sits between the two Onizuka variants, retaining the per-coordinate advantage of the horseshoe at these tails, and under Beta noise the three methods are close, with the strong skewness of the error distribution near the lower tail leaving little separation in MSE and MAD. Across all noise distributions the credible intervals produced by the Joint Sequential Sampler are substantially narrower than those of both competitors, by a factor of three to five under Gaussian and mixed normal noise. Consistent with the convention adopted for the interior quantiles, interval width is reported here only as a descriptive summary of posterior dispersion; the magnitude of this contraction makes empirical coverage the decisive diagnostic,

Table 11: Mean absolute deviation (MAD) for the piecewise constant signal at extreme quantile levels, difference order $k = 0$, across noise distributions.

	MAD			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.2139	0.1864	0.1815	0.2126
Onizuka (Laplace)	0.2979	0.2691	0.2681	0.2968
Joint Sequential Sampler	0.2523	0.1901	0.1315	0.1765
(II) Beta	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.0194	0.0107	0.1028	0.1078
Onizuka (Laplace)	0.0757	0.0539	0.1504	0.1572
Joint Sequential Sampler	0.0238	0.0178	0.0828	0.0997
(III) Mixed normal	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.5469	0.4485	0.4567	0.5561
Onizuka (Laplace)	0.6957	0.6052	0.6140	0.7041
Joint Sequential Sampler	0.5697	0.3818	0.2752	0.4404

Table 12: Mean credible interval width (MCIW) for the piecewise constant signal at extreme quantile levels, difference order $k = 0$, across noise distributions. Width is a descriptive summary of posterior dispersion and should be read alongside empirical coverage.

	MCIW			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.3938	0.4140	0.4024	0.3824
Onizuka (Laplace)	0.6919	0.6627	0.6595	0.6862
Joint Sequential Sampler	0.0167	0.0538	0.0652	0.0210
(II) Beta	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.0433	0.0363	0.2304	0.2115
Onizuka (Laplace)	0.3656	0.3084	0.4111	0.4675
Joint Sequential Sampler	0.0530	0.0235	0.0162	0.0465
(III) Mixed normal	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.8976	0.9257	0.9297	0.9064
Onizuka (Laplace)	1.2581	1.2601	1.2553	1.2487
Joint Sequential Sampler	0.0271	0.0830	0.1145	0.0474

and these widths should not be read as improved uncertainty quantification until coverage is established.

Table 13: Mean squared error (MSE) for the varying smoothness signal at extreme quantile levels, difference order $k = 1$, across noise distributions. The Joint Sequential Sampler is the sequential joint sampler equipped with sentinel quantiles.

	MSE			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.043	0.032	0.030	0.041
Onizuka (Laplace)	0.068	0.051	0.046	0.065
Joint Sequential Sampler	0.051	0.036	0.032	0.049
(II) Beta				
Onizuka (Horseshoe)	0.002	0.001	0.010	0.012
Onizuka (Laplace)	0.001	0.001	0.016	0.020
Joint Sequential Sampler	0.002	0.001	0.013	0.019
(III) Mixed normal				
Onizuka (Horseshoe)	0.197	0.132	0.144	0.212
Onizuka (Laplace)	0.317	0.221	0.238	0.336
Joint Sequential Sampler	0.197	0.128	0.124	0.192

3.2.3 Smooth Function

Tables 16, 17, and 18 report MSE, MAD, and MCIW for the smooth signal at the extreme quantile levels, fit at difference order $k = 2$. In contrast with the interior levels, where the individual sampler leads on this signal at nearly every level, the extreme quantiles are where the Onizuka variants are most competitive against the joint construction. Under Gaussian and mixed normal noise the horseshoe attains the lowest MSE at all four levels, though the margins over the Joint Sequential Sampler are modest—differences appear in the third decimal place under Gaussian noise—and the Joint Sequential Sampler improves on the Laplace variant at every level under mixed normal noise while splitting the comparison under Gaussian noise. The clearest deficit is under Beta noise, where the Joint Sequential Sampler trails both competitors at all four levels; the absolute errors are small throughout, but at the lower tail its MSE is roughly three to five times that of the Onizuka variants, and at the upper tail roughly double. A structural reading is available: the sentinel offsets are fixed constants, while under the strongly skewed Beta error the true quantile curves near the lower boundary are extremely tightly spaced, so any mismatch between the imposed sentinel spacing and the true local spacing is transmitted to the target trend through the non-crossing constraint. This pattern is consistent with that mechanism, though it has not been isolated experimentally. As on the other signals, the credible intervals of the Joint Sequential Sampler are much narrower than those of both competitors—by roughly a factor

Table 14: Mean absolute deviation (MAD) for the varying smoothness signal at extreme quantile levels, difference order $k = 1$, across noise distributions.

	MAD			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.163	0.137	0.134	0.160
Onizuka (Laplace)	0.214	0.180	0.174	0.211
Joint Sequential Sampler	0.183	0.150	0.142	0.182
(II) Beta	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.024	0.018	0.078	0.086
Onizuka (Laplace)	0.028	0.021	0.103	0.117
Joint Sequential Sampler	0.021	0.023	0.093	0.114
(III) Mixed normal	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.369	0.292	0.300	0.378
Onizuka (Laplace)	0.482	0.390	0.404	0.495
Joint Sequential Sampler	0.368	0.290	0.284	0.364

Table 15: Mean credible interval width (MCIW) for the varying smoothness signal at extreme quantile levels, difference order $k = 1$, across noise distributions. Width is a descriptive summary of posterior dispersion and should be read alongside empirical coverage.

	MCIW			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.268	0.269	0.267	0.254
Onizuka (Laplace)	0.362	0.357	0.340	0.348
Joint Sequential Sampler	0.069	0.063	0.070	0.070
(II) Beta	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.079	0.070	0.158	0.154
Onizuka (Laplace)	0.146	0.118	0.212	0.227
Joint Sequential Sampler	0.041	0.006	0.032	0.053
(III) Mixed normal	0.01	0.02	0.98	0.99
Onizuka (Horseshoe)	0.507	0.536	0.553	0.525
Onizuka (Laplace)	0.674	0.696	0.720	0.705
Joint Sequential Sampler	0.128	0.153	0.157	0.126

of four under Gaussian and mixed normal noise (Table 18)—and this contraction is addressed in the closing discussion below.

Table 16: Mean squared error (MSE) for the smooth signal at extreme quantile levels, difference order $k = 2$, across noise distributions. The Joint Sequential Sampler is the sequential joint sampler equipped with sentinel quantiles.

	MSE			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.0422	0.0310	0.0283	0.0396
Onizuka (Laplace)	0.0445	0.0347	0.0317	0.0424
Joint Sequential Sampler	0.0491	0.0319	0.0305	0.0476
(II) Beta				
Onizuka (Horseshoe)	0.0004	0.0005	0.0088	0.0114
Onizuka (Laplace)	0.0003	0.0004	0.0096	0.0115
Joint Sequential Sampler	0.0015	0.0018	0.0160	0.0231
(III) Mixed normal				
Onizuka (Horseshoe)	0.2017	0.1402	0.1445	0.2027
Onizuka (Laplace)	0.2376	0.1652	0.1726	0.2414
Joint Sequential Sampler	0.2338	0.1509	0.1486	0.2310

The interval widths of the Joint Sequential Sampler reported throughout this subsection (Tables 12, 15, and 18) should be read in light of how the inferential set is constructed rather than as a direct claim of sharper uncertainty quantification. Under the sentinel mechanism the target quantile is bracketed at every index by the innermost sentinels on either side, and the non-crossing constraint forces the target trend to remain between these neighbours throughout the chain. With the compressed spacing used near the boundary—0.003 below the target and 0.01 above it for $\tau = 0.01$, and analogously for the other extreme levels—the admissible vertical band for the target is narrow by construction, which mechanically limits the dispersion of its posterior and hence the width of its credible intervals. The small MCIW values reported in these tables are therefore largely a consequence of this geometric confinement, and are not directly comparable to the widths of the unconstrained Onizuka samplers, which estimate each quantile in isolation. They should be understood only in that sense, and empirical coverage remains the decisive measure of whether the resulting intervals are calibrated.

Table 17: Mean absolute deviation (MAD) for the smooth signal at extreme quantile levels, difference order $k = 2$, across noise distributions.

	MAD			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.1608	0.1351	0.1316	0.1590
Onizuka (Laplace)	0.1672	0.1440	0.1403	0.1660
Joint Sequential Sampler	0.1788	0.1400	0.1382	0.1780
(II) Beta				
Onizuka (Horseshoe)	0.0119	0.0096	0.0742	0.0860
Onizuka (Laplace)	0.0105	0.0090	0.0773	0.0863
Joint Sequential Sampler	0.0211	0.0231	0.1025	0.1272
(III) Mixed normal				
Onizuka (Horseshoe)	0.3698	0.3009	0.2995	0.3679
Onizuka (Laplace)	0.4035	0.3283	0.3322	0.4057
Joint Sequential Sampler	0.4043	0.3155	0.3093	0.4010

Table 18: Mean credible interval width (MCIW) for the smooth signal at extreme quantile levels, difference order $k = 2$, across noise distributions. Width is a descriptive summary of posterior dispersion and should be read alongside empirical coverage.

	MCIW			
	0.01	0.02	0.98	0.99
(I) Gauss				
Onizuka (Horseshoe)	0.2439	0.2501	0.2466	0.2415
Onizuka (Laplace)	0.2345	0.2495	0.2448	0.2374
Joint Sequential Sampler	0.0584	0.0612	0.0611	0.0584
(II) Beta				
Onizuka (Horseshoe)	0.0703	0.0599	0.1442	0.1455
Onizuka (Laplace)	0.0606	0.0529	0.1405	0.1358
Joint Sequential Sampler	0.0115	0.0024	0.0197	0.0374
(III) Mixed normal				
Onizuka (Horseshoe)	0.4815	0.5122	0.5172	0.4909
Onizuka (Laplace)	0.5027	0.5253	0.5370	0.5167
Joint Sequential Sampler	0.1074	0.1244	0.1262	0.1084

4 Discussion

Summary of findings. The simulation study establishes that the Single Quantile Sampler is a competitive Bayesian estimator for quantile trend filtering across a range of signal structures, noise distributions, and quantile levels. It consistently improves on Onizuka et al. (2024) under the Laplace prior in both MSE and MAD, and matches or approaches the horseshoe variant at interior quantile levels under all three noise distributions. The advantage of the globally penalized P-MALA approach is most pronounced on the piecewise quadratic-compatible varying smoothness signal, where the global prior’s implicit bias toward the correct order of smoothness is well matched to the signal structure. The pattern reverses, as expected, on the piecewise constant signal under Gaussian noise at interior quantiles, where the per-coefficient horseshoe prior’s local adaptivity provides a structural advantage that a globally penalized model cannot replicate. These findings are consistent with the general principle that local adaptivity benefits estimation when the target signal has genuinely heterogeneous smoothness, and that the associated computational and modeling complexity is the price paid for that benefit.

Scalability. A central motivation for the proposed approach is computational. The exact Gibbs sampler of Onizuka et al. (2024) requires, at each iteration, sampling a multivariate Gaussian whose precision matrix involves the diagonal weight vector w , which is resampled every iteration and prevents any reuse of the matrix factorization. The resulting $O(n^3)$ per-iteration cost becomes prohibitive as the sample size grows. The proposed sampler replaces this with an ADMM proximal subproblem whose coefficient matrix $\mathbf{A} = \gamma^{-1}\mathbf{I} + \rho D^\top D + \rho\mathbf{I}$ is fixed within and across Gibbs iterations. The matrix is sparse and banded with bandwidth $k + 1$, so its Cholesky factorization costs $O(n)$ and is computed once per Gibbs step, with all subsequent ADMM backsolves also $O(n)$. The per-iteration cost of the proposed sampler therefore scales linearly with n , independently of the number of ADMM steps, whereas the Onizuka exact sampler scales cubically. At the sample size $n = 400$ used in the simulation study this difference is already appreciable; at larger n it is decisive, and the proposed approach opens the door to sample sizes where the Onizuka exact sampler is simply not feasible.

Role of the scale parameter. Treating the ALD scale σ as an unknown parameter rather than fixing it at the optimization convention of unity is a substantive modeling decision with several consequences. Within the Gibbs scheme, σ and λ interact with the trend estimate in complementary ways: σ calibrates the effective likelihood temperature, controlling how sharply the gradient of the smoothed check loss responds to residuals, while λ calibrates the effective smoothness penalty. Decoupling these two sources of regularization allows each to adapt to a different aspect of the data — noise level and signal roughness respectively — and prevents the single tuning parameter of the frequentist formulation from having to absorb both roles simultaneously. Under model misspecification, which is present in all three noise distributions considered here since none is asymmetric Laplace, the posterior for σ provides a partial correction to the likelihood temperature, reducing the sensitivity of the estimates to the mismatch between the working model and the true data-generating process. A practical caveat is that σ and λ are not fully separable in their effects on the posterior for

θ : both appear in the effective curvature seen by the P-MALA proposal, and under severe misspecification their joint posterior may exhibit a soft ridge. The weakly informative prior $\sigma \sim \text{IG}(0.1, 0.1)$ is sufficient to regularize this in all settings considered, but sensitivity of the posterior to this prior choice warrants attention in applications where the noise distribution is far from symmetric or unimodal.

Limitations. The globally penalized prior is the principal structural limitation of the proposed approach. By placing a single shared penalty λ on all entries of $D\theta$, the model encodes a homogeneous prior belief about the smoothness of the trend. This is appropriate when the signal is globally smooth or when the difference order k is chosen to match the signal class, as in the varying smoothness experiments, but it is a deliberate simplification relative to the per-coefficient shrinkage of the horseshoe prior. On piecewise constant signals, the globally penalized model must use a single λ to accommodate both flat segments and jump locations, which leads to a bias-variance tradeoff that the horseshoe prior resolves more cleanly through local adaptivity. The persistent advantage of Onizuka (Horseshoe) on the PC signal across Gaussian and mixed normal noise is a direct consequence of this structural difference and is not expected to diminish with larger samples or better tuning. Users whose signals are known to be piecewise constant should weigh this tradeoff explicitly.

The sentinel strategy as a heuristic. The sentinel quantile construction introduced in Section 2.5 resolves the gradient degeneracy of P-MALA at extreme quantile levels and produces stable estimates in the simulation study. It is, however, a practical device rather than a formally justified procedure. The specific recommendations — up to five sentinels per side, equally spaced within each side, with spacings chosen to fill the available probability room — are motivated by empirical experience and by the requirement that the sentinels be close enough to the target to provide a tight dynamic bound while remaining far enough apart to avoid degeneracy among themselves.

Extensions. Several directions for extension follow naturally from the present work. The P-MALA framework accommodates non-Euclidean preconditioning through the proximal operator, and a preconditioned variant exploiting the banded structure of $D^\top D$ may improve mixing at higher difference orders without increasing per-iteration cost. The sentinel strategy is applicable to any gradient-based sampler for quantile estimation and could be combined with locally adaptive priors, potentially recovering the accuracy advantages of the horseshoe prior at extreme quantile levels while retaining the computational structure of the present approach. Extension to multivariate or spatially indexed signals requires generalization of the difference operator D to non-chain graph structures, for which the banded linear system no longer applies directly; spectral or multigrid solvers may serve a similar role in those settings. Finally, the theoretical properties of the P-MALA approximation under the non-smooth ALD likelihood — including mixing time bounds and the effect of the smoothing parameter ε on the approximation error in the acceptance ratio — remain to be established.

References

- Brantley, H. L., Guinness, J., and Chi, E. C. (2020). Baseline drift estimation for air quality data using quantile trend filtering. *The Annals of Applied Statistics*, 14(2):585–604.
- Fasiolo, M., Wood, S. N., Zaffran, M., Nedellec, R., and Goude, Y. (2021). Fast calibrated additive quantile regression. *Journal of the American Statistical Association*, 116(535):1402–1412.
- Koenker, R. and Bassett Jr, G. (1978). Regression quantiles. *Econometrica: journal of the Econometric Society*, pages 33–50.
- Madrid Padilla, O. H. and Chatterjee, S. (2022). Risk bounds for quantile trend filtering. *Biometrika*, 109(3):751–768.
- Onizuka, T., Hashimoto, S., and Sugasawa, S. (2024). Fast and locally adaptive bayesian quantile smoothing using calibrated variational approximations. *Statistics and Computing*, 34(1):15.
- Pereyra, M. (2016). Proximal markov chain monte carlo algorithms. *Statistics and Computing*, 26(4):745–760.
- Tibshirani, R. J. (2014). Adaptive piecewise polynomial estimation via trend filtering. *The Annals of Statistics*, 42(1):285–323.